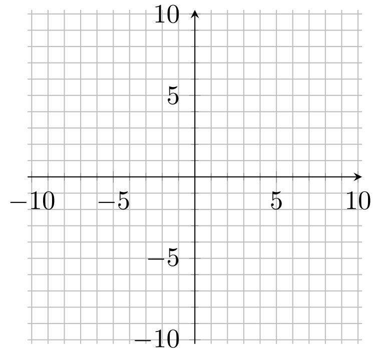
Exam 3 (make-up) Friday 1st November, 2013 **Name:**

Use the function $f(x) = (x-2)^2 - 6$ **to answer problems 1–4.**



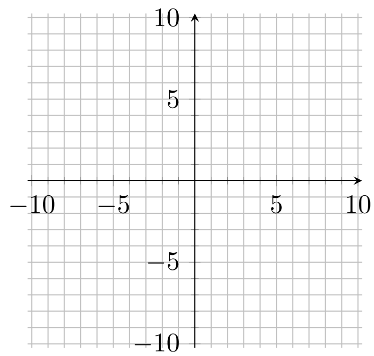
1. Find the coordinates of the vertex for the parabola defined by $f(x)$.

2. Find the axis of symmetry of the parabola defined by $f(x)$.

3. Find the range of $f(x)$.

4. Graph $f(x)$ clearly showing the vertex and y-intercept.

Use the function $g(x) = -x^2 + 4x + 4$ **to answer problems 5–8.**



5. Find the coordinates of the vertex for the parabola defined by $g(x)$.

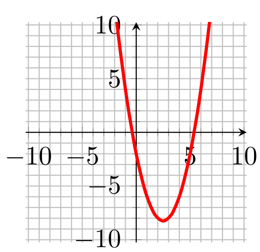
6. Find the minimum of $g(x)$.

7. Find the domain of $g(x)$.

8. Graph $g(x)$ clearly showing the vertex and y-intercept.

The graph of the function is given. Determine the function's equation.

9.



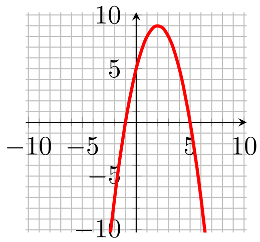
(A) $x^2 - 5x$

(B) $x^2 - 5x + 2$

(C) $x^2 - 5x - 2$

(D) $-x^2 - 5x + 2$

10.



(A) $x^2 - 4x$

(B) $x^2 + 4x + 5$

(C) $-x^2 - 4x + 5$

(D) $-x^2 + 4x + 5$

Determine if the functions in problems 11–14 are polynomials.

11. $f(x) = 3 + 12x^{3/2} + 2x$

(A) Yes **(B)** No

12. $f(x) = \pi x^4 + \sqrt{2}x^3 + x$

(A) Yes **(B)** No

13. $f(x) = x^{-2} - 2x^5 - 6$

(A) Yes **(B)** No

14. $f(x) = x + x^2 + 1$

(A) Yes **(B)** No

Use the leading coefficient test to determine the end behavior in problems 15–17.

15. $f(x) = -x^2 + 2x - 3$

(A) falls to the left and falls to the right

(B) rises to the left and falls to the right

(C) rises to the left and rises to the right

(D) falls to the left and rises to the right

16. $f(x) = -x^3 + x^4 - 3$

(A) falls to the left and falls to the right

(B) rises to the left and falls to the right

(C) rises to the left and rises to the right

(D) falls to the left and rises to the right

17. $f(x) = -x^5 + 2x^3 - 5x + 7$

(A) falls to the left and falls to the right

(B) rises to the left and falls to the right

(C) rises to the left and rises to the right

(D) falls to the left and rises to the right

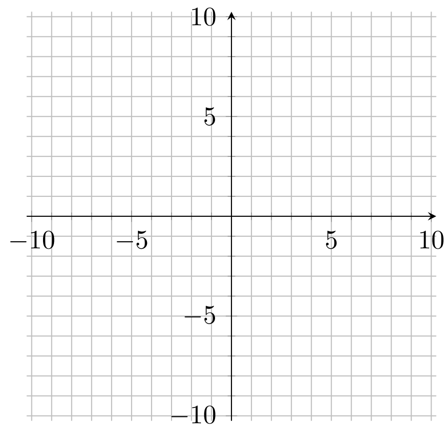
Find the degree of the polynomials in problems 11–14.

18. $f(x) = x + 3$

19. $f(x) = -x^2 + x^{10} + x$

Use the function $f(x) = \frac{1}{2}(x-2)^2(x+3)$ **to answer problems 20–22.**

20. Find the x-intercepts of $f(x)$.

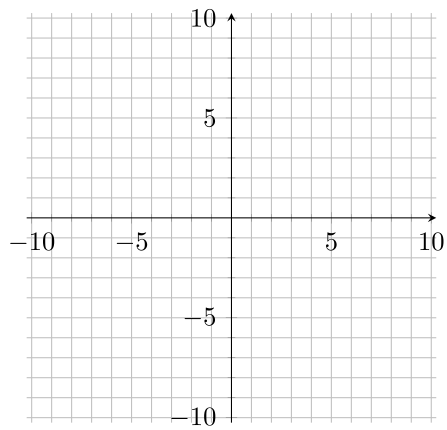


21. At each x-intercept determine whether the graph crosses the x-axis, or touches the x-axis and turns around .

22. Draw a rough sketch of $f(x)$ using your above answers.

Use the function $g(x) = x^4 - 4x^2$ **to answer problems 23–25.**

23. Find the x-intercepts of $g(x)$.



24. At each x-intercept determine whether the graph crosses the x-axis, or touches the x-axis and turns around .

25. Draw a rough sketch of $g(x)$ using your above answers.